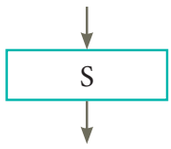
programming language. However, even pseudo code must be rigorous and correct. Pseudo code is the most widely used notation to represent algorithms.

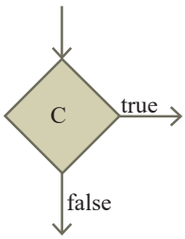
7.1.3 Flowcharts

Flowchart is a diagrammatic notation for representing algorithms. They show the control flow of algorithms using diagrams in a visual manner. In flowcharts, rectangular boxes represent simple statements, diamond-shaped boxes represent conditions, and arrows describe how the control flows during the execution of the algorithm. A flowchart is a collection of boxes containing statements and conditions which are connected by arrows showing the order in which the boxes are to be executed.

1. A statement is contained in a rectangular box with a single outgoing arrow,which points to the box to be executed next.



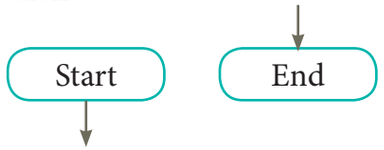
2. A condition is contained in a diamond-shaped box with two outgoing arrows, labeled true and false. The true arrow points to the box to be executed next if the condition is true, and the false arrow points to the box to be executed next if the condition is false.



3. Parallelogram boxes represent inputs given and outputs produced.



4. Special boxes marked Start and the End are used to indicate the start and the end of an execution:



The flowchart of an algorithm to compute the quotient and remainder after dividing an integer A by another integer B is shown in Figure 7.1, illustrating the different boxes such as input, output, condition, and assignment, and the control flow between the boxes. The algorithm is explained in Example 7.4.

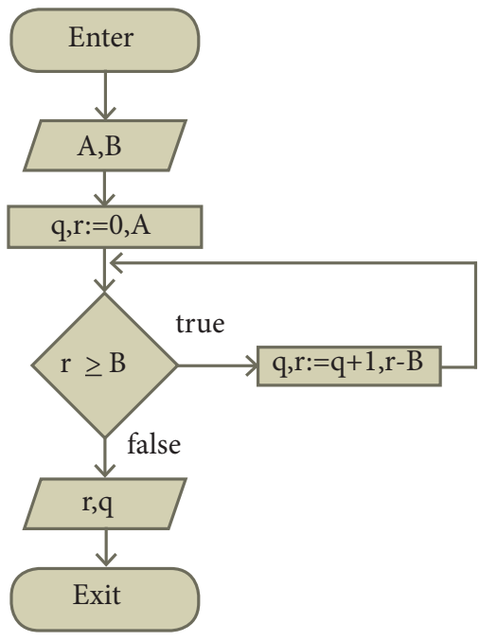


Figure 7.1: Flowchart for integer division

Flowcharts also have disadvantages. (1) Flowcharts are less compact than representation of algorithms in programming language or pseudo code. (2) They obscure the basic hierarchical structure of the algorithms. (3) Alternative statements and loops are disciplined control flow structures. Flowcharts do not restrict us to disciplined control flow structures.

Chapter 7 Page 088-101.indd 89 3/24/2020 9:14:50 AM